Ankle constraint alters cadence tracking and muscle coordination in a cadence-dependent manner: a factorial optimal-control study

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Introduction

Functional electrical stimulation (FES) cycling is a common rehabilitation modality after spinal cord injury (SCI) [1]. However, devices and models differ in whether ankle motion and ankle-muscle actuation are permitted. Musculoskeletal models often constrain the ankle for numerical convenience, although ankle muscles influence tibiofemoral loading [2], power transfer during the downstroke through hip–ankle coordination [3], and potentially proximal muscle recruitment. It remains unclear whether the effects of ankle constraint depend on cadence and ergometer geometry. We tested whether the effects of ankle constraint on cadence tracking and muscle coordination depend on cadence and ergometer geometry.

Methods

Thirty-six muscle-driven simulations were produced in OpenSim Moco (direct collocation, Hermite–Simpson, 25 mesh intervals, CasADi/IPOPT) following a full factorial design: three subject models × two ergometer geometries (MOTOmed, clinical; Berkel, performance-oriented) × three cadences (33, 50, 70 rpm) × two ankle conditions (active vs constrained), yielding 18 matched pairs. The active condition allowed ankle motion, whereas the constrained condition locked the ankle coordinate; all other model and optimisation settings were identical within matched pairs. Nine Hill-type muscle–tendon units per limb were actuated. Muscle activation was dimensionless and normalised from 0 (inactive) to 1 (fully activated). Each problem minimised cadence-tracking error, muscle effort and joint contact forces (JCF) over 2 s; all optimisations converged. All conditions were simulated using the same crank resistance of 8 Nm, allowing the isolated effects of ankle constraint, cadence and ergometer geometry to be compared. Solutions were phase-aligned to top dead centre (maximum knee flexion) and compared over one crank revolution.

Results

Constraining the ankle pinned soleus and tibialis anterior at the lower activation bound (≤0.011) in all 18 constrained simulations; active-ankle models recruited them phasically (soleus 0.020–0.065, tibialis anterior 0.020–0.087). The optimisation cost of ankle constraint increased with cadence in all six subject-ergometer combinations (Fig. 1): the median increase was 2.4% at 33 rpm, 9.3% at 50 rpm and 43.4% at 70 rpm (range 21–70%). Constrained models failed to hold the target cadence, the deficit rising from 0.2 to 7.6 rpm (max 10.7). Differences between ergometer geometries exceeded those among the three subject models: increases at 70 rpm were 54–70% for Berkel and 21–32% for MOTOmed. In contrast, knee JCF decreased in all 18 comparisons by a median of 7.6% (range 3.0–12.1%), while hip JCF changed inconsistently (median −2.4%). In active-ankle simulations, ankle range of motion was larger for the Berkel (mean 42°) than Motomed (35°).

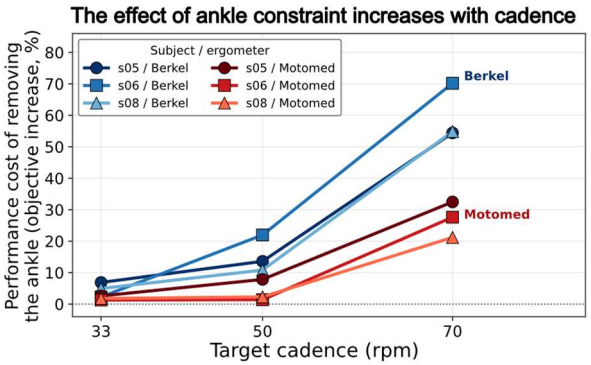


Figure 1: Relative increase in the optimal-control objective under ankle constraint versus target cadence for six subject–ergometer combinations. Blue = Berkel; red = MOTOmed. The objective increase was greater at higher cadences and for the Berkel geometry.

Conclusions

Constraining the ankle removed ankle torque generation as a coordination option, suppressed soleus and tibialis anterior recruitment, and impaired cadence tracking. These effects were strongly cadence- and geometry-dependent, consistent with an important role for ankle-mediated power transfer during the downstroke [3]. Ankle constraint should therefore not be applied as a generic modelling simplification. Its suitability depends on cadence and ergometer geometry: acceptable at low cadence, but inadequate at 70 rpm, where it altered predicted coordination and prevented accurate tracking of the target cadence. Thus, a modelling simplification with little effect at low cadence may become a major source of coordination error as cadence increases. Knee JCF also decreased, although this finding should be interpreted cautiously because constrained simulations achieved lower cadences. Although the qualitative trends were consistent, their magnitude may depend on resistance and objective-function weighting. Resistance sensitivity and EMG validation remain to be tested.

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References

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